

Global gold production grew by only 50% over the past two decades, while its real, inflation-adjusted price skyrocketed by 350%. This massive gap has triggered a modern gold rush across Africa, turning low-yield regions into highly lucrative targets. As a result, unregulated mining is expanding rapidly, putting severe environmental pressure on vital river systems, including those inside protected nature reserves.

A primary example is the Papa River, located northeast of Chinko near the South Sudanese border. A report of active gold mining from Chinko prompted this analysis. The area is exceptionally remote; the nearest camp sits fourteen hours away from any town of at least 5,000 people. Maps from the 1930s reveal that former villages, including Tidi and Ouligui, sit directly on gold-bearing rock. While no active settlements remain within 3 km of these sites today, more than twenty historic village names line the Papa River and its tributaries, where only four small camps currently exist. Our working hypothesis is that miners are returning to these long-abandoned settlement sites - possibly guided by oral history - because historic settlement on gold-bearing ground often highlights deposits that are practical to exploit. A pattern that extends across the border: the highest-ranked mining candidate in South Sudan is a former village situated on gold-bearing rock in the Busseri headwaters above Wau. Notably, four out of five settlements established since 2015 lie within 3 km of the historic villages marked on those 1930s maps. These findings identify clear priorities for field investigation; they do not, however, confirm active mining.

The extraction boom impacts the broader cross-border frontier, including poorly monitored areas spanning the sparsely inhabited wilderness between Radom, Numatina, and Southern National Park. We propose this USD 1.52 million, two-year priority intervention to secure legal and community protections before informal mining networks entrench themselves and mineral rights are locked in.

We will deploy seasonal ground teams and community wildlife agents to critical locations, working in coordination with the South Sudan Wildlife Service alongside national and regional partners. Three complementary structures will operate across the primary drainage basins:

ECHO Teams (five people: four community scouts and a leader) will provide environmental education, resource monitoring, and continuous dialogue within resident border communities. Each ECHO team is attached to one proposed Community Conservancy.

TANGO Teams (five people) will travel alongside transhumant herders, linking protected-area agreements with grazing rights, water points, veterinary care, and basic medical support. Each TANGO team is attached to one herd corridor and moves from the month the herds arrive until the end of the dry season.

Focal Points (one person each) in Zémio, and in the second year Tambura and Wau, will manage coordination with county and traditional authorities, supported centrally from Juba. Seasonal migration sets the operating calendar.

The proposed future Pongo-Wau-Numatina National Park recorded 158,844 fire detections during 2024-2025, with 99% occurring between November and February. The park's aim is to put the first real protection on the Numatina-Chelkou-Boro savannas, which WCS's 2015/16 aerial survey named a known-but-unassessed key wildlife area and which nobody has flown since 2007. The boundary as received covers 15,811 km² of Wau Wilderness ground, and on today's satellite evidence it is empty: one cattle camp of about one estimated person, no cropland, 1.5 km² of clearing in twenty-five years and none since 2020, no reported working and no modelled gold target inside it. Measured with the same yardstick as every other zone, the boundary classifies as core, park-grade land. The wildlife is still there: a 2022 field mission recorded elephant south of Deim Zubeir, hippo and crocodile in the Pongo River, and lion, leopard, buffalo and chimpanzee along the Wau-Raja axes; the park's south-eastern edge meets Southern National Park - home to the 165 Derby's eland counted in 2007, the only population west of the Nile - and 270 tracked fire fronts moved between the two shapes. Emptiness is also the legal argument: the 2026 Wildlife Act does not allow community land to be taken into a park without prior acquisition, so a boundary kept off the inhabited rim can be declared without displacing anyone. The line itself needs work before a drafter sees it. Only 35% of its perimeter runs along something a person can point at - Bo River, K. Nfotoro, Kenarofe - and 65% crosses open bush. When the whole study area is zoned by machine (see below), the park re-emerges as one 3.25 million ha block holding 10 people, bounded by CAF border, Nahr al Jur, Bo River, Tambura-Wau road, R. Pongo. The block as a whole passes the park test - 10 people, no cropland, 0.8 km² cleared since 2020, no reported working; the solver labels it wilderness rather than core only because the cap we set on new park land (6 million hectares across the whole study area) was already spent on the gazetted parks. The drawn boundary is the right idea drawn smaller than the evidence allows.

Because most fire fronts reaching the area are associated with herder movement, field teams will operate primarily from December to February. Educational outreach in villages, legal and administrative work will continue during the rainy season.

South Sudan's Mining Act 2012 allows state authorities to issue artisanal gold licenses, prohibiting mining within national parks only if explicitly stated in the park's founding instrument. The state desks concerned are those of Lakes (E4), Northern Bahr el Ghazal (T1, T2), Western Bahr el Ghazal (E5, T1, T2), Western Equatoria (E3, T3, T4, T5) - the four states whose ground the staffed zones lie on, by the OCHA COD-AB SSD v03 (2022-12-19) boundaries. To counter this vulnerability, the intervention will seek a Gazette closure order against new mineral-title applications and establish a clear boundary and explicit mining ban within the regulations of the 2026 Wildlife Act.

We will also register Community Conservancies under section 14 of the 2026 Act, each drawn to a boundary a villager can walk. The candidates were not sketched by hand (see "How the areas were drawn" below). The study area was cut into 2,130 units along rivers, khors, swamp edges, hill chains and 1930s district lines; every unit was measured with the same people, fire, cropland, clearing and gold-target layers that produced this report; and a single optimisation gave every unit one of four classes - park-grade core, wilderness, community conservancy or herd corridor - so that no two proposals overlap and each boundary is shared with its neighbour. Of the 155 community zones this produced, we take the five where people and modelled gold coincide most sharply, and station an ECHO team in each. Areas are in hectares; populations are satellite lower bounds.

Year one: Bandasi and Songo. The Bandasi conservancy (E1), 144,400 ha and 29,158 people in 13 clusters between Asa, Mbomou, secondary road, primary road in Ango (COD) 88% (Ango); Zémio (CAF) 9%, holds 11 gold-target cells and two reported workings and one settlement founded since 2015. 94% of its boundary is on named features and 95% of its perimeter shields core or corridor land; it lies in another jurisdiction (COD) and would be a partner conservancy, not ours. The Songo conservancy (E2), 134,400 ha and 37,847 people between Umbelasha, J. KOAYA - J. NOAYA 4428 - J. HELISI ridge, Adda, W. OTO DO in Al Radoum (SDN) 100%, holds 6 gold-target cells and six reported workings and one settlement founded since 2015. 66% of its boundary is on named features and 58% of its perimeter shields core or corridor land; it lies in another jurisdiction (SDN) and would be a partner conservancy, not ours.

Year two: Tambura and Cueibet and Wau. The Tambura conservancy (E3), 125,600 ha and 35,817 people between Tambura - Yubo Road, Ezo - Yubo Road, R. Yabo, R. Ybo in Tambura County 89% (Mupoi, Source Yubu, Tambura); Ezo County 9% (Bariguna), holds 8 gold-target cells, two abandoned villages on gold-bearing rock. 79% of its boundary is on named features and 49% of its perimeter shields core or corridor land. The Cueibet conservancy (E4), 105,600 ha and 43,227 people between Wahko, River Gulnam, Rumbek-Tonj Road, R.Tiju in Cueibet County 93% (Abiriu, Ngap, Duony, Malou Pec); Wulu County 7% (Bargel), holds 15 gold-target cells, one abandoned village on gold-bearing rock and six settlements founded since 2015; it ranks first of all 155 community zones on gold-rush urgency; cropland has grown from 0.0% to 2.0%. 85% of its boundary is on named features and 37% of its perimeter shields core or corridor land. The Wau conservancy (E5), 84,800 ha and 117,344 people between Nahr al Jur, Wau - Raja road, R. Asetti, Wau-Gogrial Road in Jur River County 75% (Udici, Marial Bai, Wau Bai); Wau County 25% (Besselia, Wau South, Wau North), holds 8 gold-target cells, one abandoned village on gold-bearing rock and three settlements founded since 2015. 87% of its boundary is on named features and 46% of its perimeter shields core or corridor land.

Together the five hold about 594,800 ha and 263,393 people. All are herder country - 94-99% of their fire fronts are transhumance - which is why they must be conservancies with grazing rules, not exclusion zones.

Registration will start at Bandasi and Songo. A section 14 application is made through the county, so each proposed boundary has been laid over the government administrative set (OCHA COD-AB SSD v03 (2022-12-19); the CAR, Sudan and Congo sets likewise): the staffed conservancies and corridors fall in 15 counties of Lakes, Northern Bahr el Ghazal, Western Bahr el Ghazal and Western Equatoria (Aweil Centre, Aweil West, Cueibet, Ezo, Ibba, Jur River, Maridi, Mundri West, Mvolo, Nzara, Raja, Tambura, Wau, Wulu, Yambio) and touch 45 payams, each listed with its share of the zone in the summary below, so the commissioners and payam administrators who must sign are known before the first meeting. Communities, a facilitator and legal counsel scope the applications during the first field season and file them in the 2027 rains, with the implementing partner registering in parallel with the Relief and Rehabilitation Commission. The target is at least one application accepted as complete in the first year and one conservancy authorized in the second, its community scouts registered under sections 15-16 as the legal form of the ECHO and TANGO teams. Each conservancy is framed around historic return: the traced 1930s sheets document, village by village, that these are families going back to their own ground, which the Act recognizes for long-standing residents. Between 6% and 34% of each proposed boundary still crosses ground with no river or hill to mark it, and those stretches are what the first season walks with the community. At USD 630,155 this is the largest single line in the budget.

The livestock corridors will be registered as wildlife corridors under section 9 of the same Act, which protects them and requires a funded impact plan from any infrastructure crossing them. Their outline is no longer a pair of map pins. The 13,178 long-range transhumance fronts of 2024-26 fall into fifteen origin-destination routes; for each route a least-cost path was traced and re-traced twelve times with the costs jittered, then widened until it holds at least 35% of that route's measured use, to at most 16 km either side. Fitted on 2023-24 and tested on the held-out 2025/26 season, each route's band captures 50-61% of the following year's movement where chance would give 0-14%. Three corridor zones are staffed. The Aluk (1930s village) corridor (T1), 2.58 million ha and 5,426 people along tertiary road, Low hill – J. ABU RASEIN – J. NGAYA ridge, Wau - Raja road, Raja-Kosho Road, carries four routes (2,137 fitted fronts) and 36 gold-target cells; herds enter in September. The Maridi corridor (T3), 1.23 million ha and 3,345 people along Juba-Rumbek-Bentiu Highway road, Nahr al Jur, tertiary road, Rohl, carries four routes (2,225 fitted fronts) and 9 gold-target cells; herds enter in October. The Tambura corridor (T5), 791,200 ha and 330 people along CAF border, R. Imatima, Mokoutou, primary road, carries two routes (909 fitted fronts) and 5 gold-target cells; herds enter in October. Together they hold 4.60 million ha and 9,101 people; the whole corridor class in the plan holds 15,069 people, so what is registered is a corridor through communities, and its rules must be negotiated with the same conservancies it crosses. The corridors do not follow the grazing zones drawn in the plan: the Numatina zone holds 0.3% of measured herd use. Herder leadership - seated in Wau with a delegate in Raja - has stated its conditions for respecting protected areas: designated alternative ground, water points, veterinary and medical support. Those conditions are budgeted at USD 91,100, with the first water point and veterinary campaign delivered, not promised, and negotiations held only in the December-February window when the herders are present.

The 2026 Act requires their written permission before natural resources can be licensed, giving local communities a formal veto over external exploitation.

Artisanal mining will be managed as a lawful livelihood: strictly prohibited within the proposed park and key headwaters, but licensed and supported within agreed zones outside them.

The two-year budget is fixed at USD 1.52 million (USD 495,405 / USD 1,029,216; the first six months USD 373,522).

No new organizations, vehicle fleets, aircraft, or enforcement equipment will be established. Year one fields 21 people (two ECHO teams, two TANGO teams, one focal point); year two adds 32 for 53 in all (five ECHO, five TANGO, three focal points).

Operations face strict boundaries: without written support from the Wildlife Service by 30 November 2026, the first field season will not move forward.

Furthermore, if any mineral title is issued along the proposed park rim, expansion plans will pause immediately.

Declaring the Pongo-Wau-Numatina National Park remains a sovereign, mid-term decision.

This intervention provides the essential groundwork that the decision currently lacks: an agreed boundary, a mining closure, communities with legal authority over their land, defined livestock corridors, and a reliable baseline for measuring future change.

The evidence backing this initiative is strong regarding settlement patterns, seasonal fire dynamics, and available legal mechanisms. It remains intentionally limited in three areas, which serve as the baseline data gaps the project aims to resolve: population figures are conservative minimum estimates; mining locations are predictive, modeled candidates rather than confirmed sites; and no comprehensive aerial wildlife survey has been completed since 2007.

EASY - HOW THE AREAS WERE DRAWN AND SELECTED

None of the zones in this plan were drawn by hand. They come from a four-step procedure that any reader can re-run, and every number in this document is printed by that procedure rather than typed.

Step 1 - cutting the ground into pieces. The study area, about 480,000 km² across South Sudan, the Central African Republic, Sudan and northern Congo, was divided into 2,130 units on a 2 km grid. The cuts follow only features a person can stand on and point at: rivers and khors, swamp and lake edges, hill chains, roads, international borders and the district lines of the 1930s Sudan Survey sheets. Where no such feature exists, geological contacts (the change from one rock type to another) were allowed as a last resort and are flagged in every description as "not visible on the ground - needs beacons". Each unit is at least 2,000 ha and its outer boundary carries a legibility score: the share of its length that runs along a named feature.

Step 2 - measuring every piece the same way. Each unit was measured with the same satellite layers used throughout this report: residents from the GHSL built-up rasters (a lower bound, never a census), settlements founded since 2015, fire detections and tracked fire fronts for 2024-26 and the share travelling with the herds, cropland in 2003 and 2019, verified forest clearing since 2020, and modelled gold targets and reported workings. Two further layers were fitted and tested rather than assumed. A conversion model (which land turned from bush to farm or village between 2015 and today) was trained and scored on spatially separate 20 km blocks, reaching an AUC of 0.86, meaning it ranks a converted cell above an unconverted one 86 times in 100. A herd-movement model grouped the 13,178 long-range transhumance fronts into fifteen origin-destination routes and estimated for each the ground the herds use (a Brownian-bridge utilisation distribution); fitted on 2023-24 and tested on the unseen 2025/26 season, each route's band captured 50-61% of the following year's movement where chance alone would give 0-14%.

Step 3 - assigning one role to every piece, all at once. Each unit receives exactly one of four classes - park-grade core, wilderness, community conservancy or herd corridor - from a single integer programme solved over all 2,130 units together. The classes follow fixed rules with physical meaning: core requires at most 0.02 people per km², under 0.05% cropland, under 0.5 km² of clearing per 1,000 km² since 2020 and no reported working; wilderness allows up to 0.5 people per km²; a corridor must be one continuous route from where a herd route starts to where it ends, holding at least 35% of that route's measured use; everything else is community land. The solver prefers boundaries on named features, keeps gazetted parks as core, and never puts people inside a park. Its weights were set before the run and were not adjusted afterwards. Where a genuine political choice exists - how much new park land to allow, how much of each herd route to protect - the whole problem was solved again at every setting (27 runs) and the trade-off printed: people whose land changes class, hectares of park, kilometres of boundary through open bush. The plan presented here is one row of that table, chosen with the reader able to see what the neighbouring rows would have cost.

Step 4 - placing the teams. Team sites were chosen only among existing settlements of 150 or more people (residence implies water; mapped water was re-checked and is printed for every site). For each zone, three candidate sites were shortlisted by the share of the zone's targets they can reach within four hours by car, motorbike or foot, using OpenStreetMap road travel times. Community zones were ranked for an ECHO team by gold-rush urgency (modelled gold targets, people times conversion risk, and how much park or corridor edge the zone shields); corridor zones for a TANGO team by herd use, herd use near residents, and gold targets in the band. A small open-weights language model (muse-glimmer-30b) was used at two points only: to choose among the three shortlisted sites for each team, given their measured reach, population, road and water figures and a fixed set of rules; and to draft each team's one-sentence and 120-word briefs from the stored zone descriptions. It never classified land, drew a boundary or produced a number, and its stated reason for each site choice is printed beside the choice so it can be checked.

What this procedure cannot do is also stated. Populations are satellite lower bounds. Gold targets are model predictions - the model's top 5% of cells captures 28% of the 89 known workings, a 5.7-fold improvement on chance, not a map of mines. The herd model explains where herds go along each route, not whether the network as a whole differs from where the country burns. And the boundaries proposed are candidates for a community to walk, not lines to be gazetted as drawn.

EASY - WHAT THE MAP SHOWS

The map covers the study area from Radom in Sudan to Garamba in Congo. Faint red hairlines are individual fire fronts tracked from satellite in 2024-26; where they bunch into dark red bands, herds are moving. Orange dots are today's towns and villages, sized by population. Olive squares are the gold model's most likely target cells (targets, not mines), triangles are places where imagery shows working-like ground, and diamonds are workings reported by others. Thin dashed lines are the machine zoning of the whole area - green for park-grade core, grey-green for wilderness, purple for community conservancy, red for herd corridor - with the boundaries drawn by the plan's authors in grey behind them, and Southern National Park tinted as the one gazetted park. The solid, filled shapes are the only places where this intervention puts people: purple blocks are the five community conservancies served by an ECHO team, red blocks are the three herd corridors served by a TANGO team, and each carries its area and population. Teal symbols are the team sites: a star for a focal point, a square for an ECHO team, a triangle for a TANGO team; filled symbols start in year one, hollow ones join in year two. Each label gives the team code and the settlement it is based in. The legend also states what the plan does not reach: 119,001 people in 12 towns of 2,000 or more lie further than 40 km from every team site.

EASY - PROPOSED ZONES AND LOCATIONS (SUMMARY)

Whole-area zoning (one solution over 2,130 units; areas as solved)

class	area km2	people	zones	note
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core (park-grade)	108,148	911	84	incl. gazetted parks
wilderness	128,620	932	36	buffer <=15 km round core
community conservancy	183,608	2,615,745	32	everyone else's land
herd corridor	61,220	15,069	21	15 routes, walkable
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Pongo-Wau-Numatina block	32,456	10	1	passes park test; see text

Community conservancies with an ECHO team (Wildlife Act 2026 s.14)

code	base	yr	country	area ha	people	gold	boundary (named)
E1	Bandasi	1	COD	144,400	29,158	11*	Asa, Mbomou, secondary road
E2	Songo	1	SDN	134,400	37,847	6*	Umbelasha, J. KOAYA – J. NOAYA 4428 – J. HELISI ridge, Adda
E3	Tambura	2	SSD	125,600	35,817	8	Tambura - Yubo Road, Ezo - Yubo Road, R. Yabo
E4	Cueibet	2	SSD	105,600	43,227	15	Wahko, River Gulnam, Rumbek-Tonj Road
E5	Wau	2	SSD	84,800	117,344	8	Nahr al Jur, Wau - Raja road, R. Asetti

total 594,800 263,393 48

gold = model top-5% target cells; * = also reported workings (Bandasi 2, Songo 6)

Herd corridors with a TANGO team (Wildlife Act 2026 s.9)

code	base	yr	country	area ha	people	routes	herds	from	gold	key features
T1	Aluk (193	1	SSD	2,584,000	5,426	4	2,137	Sep	36	tertiary road, Low hill – J. ABU RASEIN – J. NGAYA ridge, Wau - Raja road
T2	Busseri	1	SSD	(same corridor as T1; second team, >50 km from the first)						
T3	Maridi	2	SSD	1,225,600	3,345	4	2,225	Oct	9	Juba-Rumbek-Bentiu Highway road, Nahr al Jur, tertiary road
T4	Makpandu	2	SSD	(same corridor as T3; second team, >50 km from the first)						
T5	Tambura	2	CAF	791,200	330	2	909	Oct	5	CAF border, R. Imatima, Mokoutou

total 4,600,800 9,101

herds = fitted long-range transhumance fronts 2024-26; from = month the herds arrive

Jurisdiction of the staffed zones (OCHA COD-AB, all four countries; share of zone area; third-level units $\geq 1\%$)

code	county / state (country)	share	payams (SSD) / secteurs (COD), share
E1	Ango, Bas-Uele (COD)	88%	Ango 88%
	Zémio, Haut-Mbomou (CAF)	9%	-
E2	Al Radoum, South Darfur (SDN)	100%	-
E3	Tambura, Western Equatoria (SSD)	89%	Mupoi 56%, Source Yubu 24%, Tambura 9%
	Ezo, Western Equatoria (SSD)	9%	Bariguna 8%
E4	Cueibet, Lakes (SSD)	93%	Abiriu 39%, Ngap 25%, Duony 17%, Malou Pec 13%
	Wulu, Lakes (SSD)	7%	Bargel 7%
E5	Jur River, Western Bahr el Ghazal (SSD)	75%	Udici 41%, Marial Bai 23%, Wau Bai 11%
	Wau, Western Bahr el Ghazal (SSD)	25%	Besselia 19%, Wau South 3%, Wau North 3%
T1	Raja, Western Bahr el Ghazal (SSD)	54%	Uyujuku 33%, Ringi 13%, Raga 8%
	Wau, Western Bahr el Ghazal (SSD)	22%	Bagari 10%, Besselia 10%, Kpaile 1%
	Aweil Centre, Northern Bahr el Ghazal (SSD)	11%	Nyalath 5%, Aroyo 3%, Abul 1%, Bahr Mayen 1%
	Jur River, Western Bahr el Ghazal (SSD)	10%	Kuajiena 4%, Kangi 3%, Rocrocdong 2%, Udici 1%
	Aweil West, Northern Bahr el Ghazal (SSD)	3%	Mariem West 2%
	Al Radoum, South Darfur (SDN)	1%	-
T3	Yambio, Western Equatoria (SSD)	21%	Ri-rangu 9%, Bangasu 8%, Nadiangere 4%
	Maridi, Western Equatoria (SSD)	18%	Kozi 16%, Mambe 2%
	Ezo, Western Equatoria (SSD)	16%	Yangiri 11%, Bagidi 3%, Naandi 2%
	Ibba, Western Equatoria (SSD)	15%	Manikakara 9%, Maruko 6%
	Mundri West, Western Equatoria (SSD)	10%	Mundri 8%, Kotobi 2%
	Mvolo, Western Equatoria (SSD)	9%	Yeri 5%, Bagori 2%, Mvolo 1%
	Nzara, Western Equatoria (SSD)	7%	Ringasi 7%
	Tambura, Western Equatoria (SSD)	4%	Mupoi 4%
T5	Tambura, Western Equatoria (SSD)	37%	Namutina 28%, Tambura 9%
	Bambouti, Haut-Mbomou (CAF)	28%	-
	Obo, Haut-Mbomou (CAF)	23%	-
	Djémah, Haut-Mbomou (CAF)	11%	-

A unit below 1% of a zone is not listed. Payam shares are of the whole zone, so a county's payams sum to its share. CAF sous-préfectures and Sudan localities are the deepest official level (no payam equivalent). Where neighbouring countries' official boundaries overlap along a border the shares are rescaled to 100%.

Focal points (one person each)

code	town	yr	people	serves
F1	Zémio	1	29,065	E1
F2	Tambura	2	35,520	E3, T5
F3	Wau	2	119,818	E5, T1, T2

Staffing

year 1: 2 ECHO + 2 TANGO + 1 FP = 21 people
year 2: 5 ECHO + 5 TANGO + 3 FP = 53 people (+32)

Budget (2 years, USD)

year 1 495,405 year 2 1,029,216 total 1,524,621 first six months 373,522
see BUDGET_EASY_*.txt / .xlsx (same generator)

Sources: data/plan_zones/solver/{SOLVE,ZONES,RANK,DEPLOY}.txt via facts.json, 2026-09-07.
People are GHSL satellite lower bounds. Gold cells are model targets, not confirmed mines.
Sheet: EASY deployment map (reports/DEPLOY_MAP_*.png).